II. Short Answer Questions

1. Define centre of mass.
2. Find out the centre of mass for the given geometrical structures.
   a) Equilateral triangle
   b) Cylinder c) Square
3. Define torque and mention its unit.
4. What are the conditions in which force can not produce torque?
5. Give any two examples of torque in day-to-day life.
6. What is the relation between torque and angular momentum?
7. What is equilibrium?
8. How do you distinguish between stable and unstable equilibrium?
9. Define couple.
10. State principle of moments.
11. Define centre of gravity.
12. Mention any two physical significance of moment of inertia
13. What is radius of gyration?
14. State conservation of angular momentum.
15. What are the rotational equivalents for the physical quantities, (i) mass and (ii) force?
16. What is the condition for pure rolling?
17. What is the difference between sliding and slipping?

III. Long Answer Questions

1. Explain the types of equilibrium with suitable examples.
2. Explain the method to find the centre of gravity of a irregularly shaped lamina.
3. Explain why a cyclist bends while negotiating a curve road? Arrive at the expression for angle of bending for a given velocity.
4. Derive the expression for moment of inertia of a rod about its centre and perpendicular to the rod.
5. Derive the expression for moment of inertia of a uniform ring about an axis passing through the centre and perpendicular to the plane.
6. Derive the expression for moment of inertia of a uniform disc about an axis passing through the centre and perpendicular to the plane.
7. Discuss conservation of angular momentum with example.
8. State and prove parallel axis theorem.
9. State and prove perpendicular axis theorem.
10. Discuss rolling on inclined plane and arrive at the expression for the acceleration.

IV. Numerical Problems

1. A uniform disc of mass 100g has a diameter of 10 cm. Calculate the total energy of the disc when rolling along a horizontal table with a velocity of 20 cms^{-1}. (take the surface of table as reference)

   Ans: 0.005 J

2. A particle of mass 5 units is moving with a uniform speed of $v = 3\sqrt{2}$ units in the XOY plane along the line y = x + 4. Find the magnitude of angular momentum.

   Ans: 60 units

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